

Tareq Aziz

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RESEARCH INTERESTS

Medical Imaging, Health informatics and Healthcare data science, CVD risk prediction, MRI reconstruction, and explainable AI, with a focus on developing data-driven, robust, and deployable AI solutions for healthcare, particularly in low-resource settings.

EDUCATION

M.Sc. in Biomedical Engineering (Ongoing), Islamic University, Kushtia, Bangladesh Oct 2025 – Present

B.Sc. in Biomedical Engineering, Islamic University, Kushtia, Bangladesh Jan 2020 – Oct 2024

- CGPA: 3.12 / 4.00
- Undergraduate Project: Hybrid Machine Learning–Deep Learning Framework for Early Detection of Chronic Kidney Disease using Structured Clinical Dataset.

RESEARCH EXPERIENCE

Research Engineer, AIMS Lab, [IRIIC](#), UIU July 2026– Present

Project: An Artificial Intelligence Enabled Cardiovascular Disease (CVD) Risk Prediction in Bangladesh

- Developing an AI-enabled risk prediction framework for cardiovascular disease (myocardial infarction and stroke) for the Bangladeshi population, using routinely available clinical and non-clinical features.
- Reviewed and compared established risk models (QRISK3, SCORE2, ASCVD, CVDPORT) to assess their applicability to South Asian populations.
- Conducted statistical analysis of community health data, addressing severe class imbalance with SMOTE and stratified cross-validation.

Research Assistant, Advanced Biomedical Image & Signal Analytics Lab, Islamic University Jan 2026 – Present

- Research on computer vision, MRI reconstruction, machine learning, and deep learning for medical imaging

Research Assistant, Bioimaging Research Laboratory, Islamic University Nov 2024 – July 2026

- Research on machine learning, deep learning, image processing, signal processing, and medical data analysis; optimizing deep learning techniques for real-time medical image processing.

HONORS AND AWARDS

Best Research Paper Award, IEEE SPICSCON 2025 22 November, 2025

First Runner-Up, Responsible AI Summit Poster Presentation 18 January, 2026

Top 10 Position, IEEE CS BDC Summer Symposium 2025

PUBLICATIONS

- [1] M. E. Hossen, S. M. Mahim, **T. Aziz**, F. Ahmmed, M. K. Islam, M. S. Miah. “**HybridCCT-TB: A Dual-Stream Attention Augmented Hybrid Architecture with Explainable AI for Tuberculosis Screening in Chest Radiographs**” 2025 IEEE Int’l Conf. on Signal Processing, Information, Communication and Systems (SPICSCON). ★ **Best Paper Award**. [10.1109/SPICSCON69221.2025.11503935](https://doi.org/10.1109/SPICSCON69221.2025.11503935)
- [2] M. E. Hossen, S. M. Mahim, **T. Aziz**, M. A. Tuta, F. Ahmmed, M. Alamin. “**Integrating Spatial-Channel Mixing and Attention Gating for Fine-Grained WBC Boundary Segmentation in Microscopic Hematological Scans**” 2025 IEEE Int’l Conf. on Biomedical Engineering, Computer and Information Technology for Health (BECITHCON). [10.1109/BECITHCON69222.2025.11504204](https://doi.org/10.1109/BECITHCON69222.2025.11504204)
- [3] M. E. Hossen, S. M. Mahim, M. A. Tuta, **T. Aziz**, N. J. Ahmad, A. A. Hasan Rofik, M. A. R. Rabby, F. Ahmmed, A. Islam. “**ACCT-UNet: Multi-Scale Attention and Transformer-Based Context Modeling for Efficient and Accurate Brain Tumor Segmentation in MRI**” 2026 IEEE 2nd Int’l Conf. on Quantum Photonics, Artificial Intelligence & Networking (QPAIN). [10.1109/QPAIN69676.2026.11545924](https://doi.org/10.1109/QPAIN69676.2026.11545924)
- [4] F. Ahmmed, M. E. Hossen, S. M. Mahim, M. A. R. Rabby, **T. Aziz**, N. J. Ahmad. “**Revolutionizing Colon Cancer Diagnosis: A Novel Vision Transformer and Bi-LSTM-Based Deep Learning Model**” 2026 IEEE 2nd Int’l Conf. on Quantum Photonics, Artificial Intelligence & Networking (QPAIN). [10.1109/QPAIN69676.2026.11546160](https://doi.org/10.1109/QPAIN69676.2026.11546160)

- [5] F. Ahmmed, M. E. Hossen, S. M. Mahim, **T. Aziz**, M. A. R. Rabby, N. J. Ahmad. “**A Novel CNN-ViT-LSTM Model for Bridging Temporal and Spatial Features in Brain Tumor Diagnosis Using MRI Imaging**” 2026 IEEE 2nd Int’l Conf. on Quantum Photonics, Artificial Intelligence & Networking (QPAIN). [10.1109/QPAIN69676.2026.11546615](https://doi.org/10.1109/QPAIN69676.2026.11546615)

TEACHING EXPERIENCE

Founder and Instructor , Battles Of Physics (ED-Tech) 2021-Present

- Six years of teaching experience, both online and offline; founded and run an education-technology initiative for physics learners.

Co-founder and Instructor, Artist’s Science

An Ed Tech platform For university Undergraduate Student

May 2020–Dec2024

TRAINING

Field Work on Medical Imaging Techniques (MRI, CT Scan, X-Ray & Ultrasonography) Jan–Feb 2026

Sono Diagnostic Centre Ltd, Kushtia-7000, Khulna, Bangladesh

- Completed field work (27 Jan – 3 Feb 2026) on medical imaging techniques: MRI, CT scan, X-ray, and ultrasonography.

Field Study on Nuclear Medicine & Therapeutic Technologies

Sep 2024

Institute of Nuclear Medical Physics (INMP), Bangladesh Atomic Energy Commission (BAEC)

- Completed a 5-day field study (22–26 Sep 2024) covering PET-CT, Radiotherapy (CT simulation, Treatment Planning System, Dosimetry), LINAC (Varian Clinac-iX) console operation, patient setup, and basics of Ultrasonography.

TECHNOLOGIES

Programming Languages: Python, C

ML / DL Frameworks: TensorFlow, Keras, PyTorch, scikit-learn, XGBoost

Data Science Tools: Pandas, NumPy, SciPy

Data Visualization: Matplotlib, Seaborn

Database: MySQL

Version Control: Git, GitHub, GitLab

Research Writing: LaTeX, Overleaf, Zotero, draw.io

EXTRACURRICULAR ACTIVITIES

- **Founding Member**, Islamic University Science Club
- **Former Member**, IEEE Islamic University Student Branch

REFERENCES

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